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Studierichting: Informatica

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$$f(x, y) = \begin{cases} \frac{x^2 y^2 (x + y)}{x^4 + y^4} & \text{als } (x, y) \neq (0, 0) \\ 0 & \text{als } (x, y) = (0, 0) \end{cases}$$

Continuïteit:

We weten dat:

$$(x^2 + y^2)^2 = x^4 + y^4 + 2x^2 y^2 \geq 0 \Rightarrow x^4 + y^4 \geq -2x^2 y^2$$

$$(x^2 - y^2)^2 = x^4 + y^4 - 2x^2 y^2 \geq 0 \Rightarrow x^4 + y^4 \geq 2x^2 y^2$$

Dus:

$$\begin{aligned} x^4 + y^4 &\geq |2x^2 y^2| \\ \Rightarrow \frac{1}{x^4 + y^4} &\leq \frac{1}{|2x^2 y^2|} \\ \Rightarrow \frac{x^2 y^2 |x + y|}{x^4 + y^4} &\leq \frac{x^2 y^2 |x + y|}{2x^2 y^2} \\ \Rightarrow \left| \frac{x^2 y^2 (x + y)}{x^4 + y^4} \right| &\leq \frac{|x + y|}{2} \\ \Rightarrow -\frac{|x + y|}{2} &\leq \frac{x^2 y^2 (x + y)}{x^4 + y^4} \leq \frac{|x + y|}{2} \\ \Rightarrow \lim_{(x, y) \rightarrow (0, 0)} -\frac{|x + y|}{2} &\leq \lim_{(x, y) \rightarrow (0, 0)} \frac{x^2 y^2 (x + y)}{x^4 + y^4} \leq \lim_{(x, y) \rightarrow (0, 0)} \frac{|x + y|}{2} \\ \Rightarrow 0 &\leq \lim_{(x, y) \rightarrow (0, 0)} f(x, y) \leq 0 \end{aligned}$$

$$\text{Dus } \lim_{(x, y) \rightarrow (0, 0)} f(x, y) = f(0, 0) = 0$$

$\Rightarrow f$ is continu in $(0, 0)$.

Partiële afgeleiden:

$$\begin{cases} \frac{\partial f}{\partial x}(0, 0) = \lim_{\lambda \rightarrow 0} \frac{f(\lambda, 0) - f(0, 0)}{\lambda} = \lim_{\lambda \rightarrow 0} \frac{0}{\lambda} = 0 \\ \frac{\partial f}{\partial y}(0, 0) = \lim_{\lambda \rightarrow 0} \frac{f(0, \lambda) - f(0, 0)}{\lambda} = \lim_{\lambda \rightarrow 0} \frac{0}{\lambda} = 0 \end{cases}$$

$\Rightarrow f$ is partieel afleidbaar in $(0, 0)$

Afleidbaarheid:

Stel $\bar{h} = (h_1, h_2)$ met $h_1 \neq 0$ en $h_2 \neq 0$

$$\begin{aligned} Df(\bar{0}, \bar{h}) &= \lim_{\lambda \rightarrow 0} \frac{f(\lambda h_1, \lambda h_2) - f(0, 0)}{\lambda} \\ &= \lim_{\lambda \rightarrow 0} \frac{\frac{\lambda^4 h_1^2 h_2^2 (\lambda h_1 + \lambda h_2)}{\lambda^4 h_1^4 + \lambda^4 h_2^4}}{\lambda} \\ &= \lim_{\lambda \rightarrow 0} \frac{\lambda^5 h_1^2 h_2^2 (h_1 + h_2)}{\lambda^5 (h_1^4 + h_2^4)} \\ &= \frac{h_1^2 h_2^2 (h_1 + h_2)}{h_1^4 + h_2^4} \end{aligned}$$

Limiet bestaat, dus f is afleidbaar in $(0, 0)$.

Differentieerbaarheid:

Als f differentieerbaar is, dan moet de volgende kloppen:

$$\begin{aligned} Df(\bar{0}, \bar{h}) &= Df(\bar{0}) \cdot h \\ \frac{h_1^2 h_2^2 (h_1 + h_2)}{h_1^4 + h_2^4} &= \begin{pmatrix} 0 & 0 \end{pmatrix} \cdot \begin{pmatrix} h_1 \\ h_2 \end{pmatrix} = 0 \end{aligned}$$

Dat klopt niet want h_1 en h_2 kunnen alles zijn. Dus f is niet differentieerbaar.

References